Lee Gabriel Isidro

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SUMMARY

Fourth-year Computer Science student and sole developer of VoteX, a deployed full-stack, blockchain-backed election platform built with React, Node.js, and Supabase/PostgreSQL. Builds end-to-end machine learning pipelines with XGBoost, MLflow, and FastAPI. Interested in software development, applied machine learning, and MLOps.

EDUCATION

Bataan Peninsula State University — Balanga, Bataan
Bachelor of Science in Computer Science | GWA: 1.4 (1.0 highest) | Dean's Lister — Expected July 2027

- **Thesis:** VoteX: Blockchain-Based Voting System for Bataan Christian College Inc. using Sentiment Analysis
- Relevant coursework: Data Structures, Database Systems, Software Engineering

Bataan National High School — Balanga, Bataan
Senior High School Diploma, STEM, With High Honors — 2023

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Solidity
Web & Backend: React, Node.js, FastAPI, Tailwind, REST APIs, Supabase, PostgreSQL
ML & Data: scikit-learn, XGBoost, pandas, NumPy, MLflow, YOLOv8
DevOps & Deployment: Docker, Git/GitHub, Vercel, Render, Ethereum Sepolia testnet
AI-Assisted Development: Claude, OpenAI, GitHub Copilot, DeepSeek
Interests: AI, Machine Learning, Deep Learning, Software Development, MLOps
Spoken Languages: English, Filipino

PROJECTS

VoteX: Blockchain-Based Voting System for Bataan Christian College Inc. using Sentiment Analysis (Thesis) — Jan 2026 – Sept 2026
Sole developer on a team of 3 | React, Node.js, Supabase/PostgreSQL, Solidity — vote-x.app

- Designed, built, and deployed a full-stack election platform covering setup, voter and candidate management, scheduling, voting, and live results
- Implemented role-based access for admins, voters, and candidates, with an audit log on every admin action
- Designed the database schema and migrations in Supabase/PostgreSQL
- Wrote a Solidity smart contract that publishes a hash of each ballot to the Ethereum Sepolia testnet, making votes tamper-evident and verifiable by receipt without revealing voter choices
- Built a background sentiment pipeline that scores voter comments with GPT-OSS 20B (via Groq), with automatic fallback to DeepSeek
- Evaluated by 110 users (100 students, 10 IT professionals) with an overall mean of 4.06/5 ("Very Acceptable")

Credit Card Fraud Detection Pipeline — Nov 2026
Python, XGBoost, scikit-learn, MLflow, FastAPI, Docker — github.com/gableee/fraud-detection-mlops

- Built an end-to-end ML pipeline to flag fraudulent card transactions in a highly imbalanced dataset (492 frauds in 284,807 transactions, 0.17%)
- Optimized for AUPRC instead of accuracy, since a model that always predicts "legit" is 99.83% accurate and catches no fraud
- Tuned XGBoost with a grid search and 5-fold cross-validation (40 fits), raising test AUPRC from 0.822 to 0.870
- Chose the decision threshold from out-of-fold predictions to hit a target recall, catching 86 of 98 frauds in a held-out set of 56,962 transactions
- Tracked experiments in MLflow and served the model through a FastAPI endpoint packaged in Docker

Eryysyn: AI Knowledge and Research Platform (In Progress) — Nov 2026 – Present
AI engineering project — github.com/gableee/eryysyn

- Building a platform that answers questions grounded in the user's own documents, with citations
- Planned in nine phases: full-stack foundation, LLM integration, RAG built from scratch, evaluation, LangChain comparison, web research, LangGraph agents, and production MLOps

Other Projects

- **Studystreak** (React, Python, Sept–Nov 2025): course prototype that turns uploaded PDFs, slide decks, and documents into AI-generated summaries, quizzes, and flashcards, with a Pomodoro timer
- **PracticeYOLO** (Python, 2026): counted cars in video with YOLOv8n and cvzone
- **Josephus Calculator** (Java, 2025): final project for an algorithms course

HACKATHONS

City of Balanga Kahanga-Hackathon Balanga City, Philippines
Front-end Developer | Qualifier Entry level of 22 teams Oct 2025

- Built LIWANAG (Lighting Intelligence with Automated Navigation for Analytics and Governance), a data-driven smart lighting system for public spaces
- Developed the React front end for the admin panel, dashboard analytics, and real-time IoT device status monitoring

SiKAPTala National CS & IT Competition | Team Member | *Top 10 Finalist, 600+ students* Online (Philippines), May 2026

- Collaborated with a team to enhance an existing prototype of AGAPAY, a mobile application that helps people on medication stay organized, track their daily health, and keep their doctors and family updated
- Contributed to brainstorming and feature planning to improve the prototype

LEADERSHIP EXPERIENCE

Angatleta Sports Bataan
Communication Director Aug 2023 – Aug 2024

- Led communications on the organization's Facebook page (3.5k followers), covering team news, events, and programs
- Handled communications for high-profile events, including an exhibition game against Ateneo, a Volleyball All-Stars match (UAAP players vs. Angatleta), and a Manila training camp with UAAP teams from Ateneo, Adamson, NU, UST, and DLSU
- Coordinated with more than 50 team members on announcements and event schedules

CERTIFICATIONS

AI & Machine Learning (DeepLearning.AI, Coursera): Machine Learning Specialization; Neural Networks and Deep Learning; Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

Blockchain & Security: Blockchain Specialization (University at Buffalo, Coursera); Cybersecurity, IT Specialist (Certiport); Cyber Threat Management (Cisco)

Development & Digital: Software Development, IT Specialist (Certiport); Responsive Web Design (freeCodeCamp); Google Prompting Essentials (Google, Coursera); IC3 Digital Literacy Level 2 (American Council on Education)